The Most General Quadratic Lagrangian with Torsionful Curvature, Torsion, and Electromagnetic Field Strength in 4D

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Abstract

We construct the most general Lagrangian up to second order in the Riemann–Cartan curvature $\tilde{R}_{\mu\nu\rho\sigma}$, torsion $T^{\lambda}{}_{\mu\nu}$, and electromagnetic field strength $F_{\mu\nu}$ in four-dimensional spacetime. Using `MakeContractionAnsatz` from the `xTras` package for `xAct` (Mathematica), we automatically enumerate all independent scalar invariants with guaranteed completeness.

The core result is a Lagrangian with **35 independent coupling constants**: 3 linear (Einstein–Hilbert, cosmological constant, Barbero–Immirzi), 11 parity-even quadratic, and 20 physical parity-odd quadratic (one topological term excluded). Beyond the core, ghost-free derivative extensions add up to 29 parity-even and 80 parity-odd terms (upper bounds before Bianchi reduction), and cubic terms relevant for linearization around a magnetic background add a further ∼65 safe terms. All are presented with explicit notation and classified by ghost status, parity, and mixing channel.

Contents

1 Introduction

Poincaré gauge theory (PGT) is the natural gauge-theoretic extension of general relativity that includes torsion as a dynamical field [1, 2, 6]. The most general PGT action consistent with power-counting renormalizability is quadratic in the field strengths (curvature and torsion), analogous to the Yang–Mills action for internal gauge theories.

This document enumerates the complete set of independent scalar invariants available for the PGT+electromagnetism Lagrangian in 4D, including:

- parity-odd terms (which Barker et al. [1] omit "only out of simplicity"),
- one-derivative extensions beyond the standard Yang–Mills PGT (including Barker's eWGT completion),
- cubic terms that contribute at quadratic order when linearizing around a background magnetic field (relevant for Gertsenshtein-type graviton–photon–torsion conversion),
- Chern–Simons torsion–EM couplings involving the gauge potential A_μ.

The enumeration is performed automatically using the `AllContractions` function from the `xTras` package [9] for `xAct`, which guarantees completeness by enumerating all valid index permutations modulo tensor symmetries via `xPerm` (see Appendix A).

Table 2 on page 9 provides the complete summary of all sectors with parameter counts and notation.

2 Conventions and Definitions

We work in 4D spacetime with metric signature $(-, +, +, +)$.

2.1 Riemann–Cartan Geometry

The metric-affine connection decomposes as $\Gamma^\lambda{}_{\mu\nu} = \bar{\Gamma}^\lambda{}_{\mu\nu} + K^\lambda{}_{\mu\nu}$, where $\bar{\Gamma}$ is the Levi-Civita connection and $K^\lambda{}_{\mu\nu} = \frac{1}{2}(T^\lambda{}_{\mu\nu} + T_\mu{}^\lambda{}_\nu + T_\nu{}^\lambda{}_\mu)$ is the contortion tensor.

2.2 Torsion and Its Irreducible Decomposition

The torsion tensor $T^\lambda{}_{\mu\nu} \equiv 2\,\Gamma^\lambda{}_{[\mu\nu]}$ is antisymmetric in its last two indices. It decomposes into three irreducible parts under the Lorentz group [2]:

(i) **Trace vector** (4 components): $T_\mu \equiv T^\lambda{}_{\lambda\mu}$.

(ii) **Axial pseudovector** (4 components): $S^\mu \equiv \varepsilon^{\alpha\beta\gamma\mu}\, T_{\alpha\beta\gamma}$.

(iii) **Tensor part** (16 components, traceless): $q_{\alpha\beta\gamma}$, with $q^\lambda{}_{\lambda\gamma} = 0$ and $\varepsilon^{\alpha\beta\gamma\delta}\, q_{\alpha\beta\gamma} = 0$.

The reconstruction formula is:

$$T_{\alpha\beta\mu} = \tfrac{1}{3}(T_\beta\, g_{\alpha\mu} - T_\mu\, g_{\alpha\beta}) - \tfrac{1}{6}\,\varepsilon_{\alpha\beta\mu\nu}\, S^\nu + q_{\alpha\beta\mu}\,. \quad (1)$$

2.3 Torsionful Curvature

The Riemann–Cartan curvature tensor

$$\tilde{R}^{\rho}{}_{\sigma\mu\nu} = 2\,\partial_{[\mu}\Gamma^{\rho}{}_{|\nu]\sigma} + 2\,\Gamma^{\rho}{}_{[\mu|\lambda}\,\Gamma^{\lambda}{}_{|\nu]\sigma} \tag{2}$$

is antisymmetric in each index pair but has **no pair-exchange symmetry** in general: $\tilde{R}_{\rho\sigma\mu\nu} \neq \tilde{R}_{\mu\nu\rho\sigma}$. This means the torsionful Riemann has more independent contractions than the torsion-free Riemann tensor.

The Riemann–Cartan Ricci scalar decomposes as [2]:

$$\tilde{\mathcal{R}} = R - 2\,\nabla_\alpha T^\alpha - \tfrac{4}{3}\,T_\alpha T^\alpha + \tfrac{1}{2}\,q_{\alpha\beta\gamma}\,q^{\alpha\beta\gamma} + \tfrac{1}{24}\,S_\alpha S^\alpha\,, \tag{3}$$

where R is the Levi-Civita Ricci scalar and $\nabla_\alpha T^\alpha$ is a total derivative.

2.4 Electromagnetic Field Strength

$F_{\mu\nu} = 2\,\partial_{[\mu}A_{\nu]}$, antisymmetric.

3 The General Lagrangian

The most general Lagrangian density takes the form

$$\boxed{\mathcal{L} = \mathcal{L}_{\rm lin} + \mathcal{L}_{\tilde{R}^2} + \mathcal{L}_{T^2} + \mathcal{L}_{F^2} + \mathcal{L}_{\tilde{R}F} + \mathcal{L}_{\rm odd} + \mathcal{L}_{\nabla} + \mathcal{L}_{\rm cubic} + \mathcal{L}_{\rm CS}} \tag{4}$$

where each term is detailed in the following subsections. All terms and their parameter counts are summarised in Table 2.

3.1 Linear Sector (3 parameters)

$$\mathcal{L}_{\rm lin} = -\frac{M_{\rm Pl}^2}{2}\,\tilde{\mathcal{R}} - M_{\rm Pl}^2\,\Lambda + \frac{M_{\rm Pl}^2}{2\gamma_{\rm BI}}\,\varepsilon^{\mu\nu\rho\sigma}\,\tilde{R}_{\mu\nu\rho\sigma}\,. \tag{5}$$

The third term is the *Holst term* [7] with Barbero–Immirzi parameter $\gamma_{\rm BI}$. It is topological in the torsion-free case but *dynamical with torsion*. Via the Nieh–Yan identity [8]

$$\varepsilon^{\mu\nu\rho\sigma}\big(T^{\alpha}{}_{\mu\nu}\,T_{\alpha\rho\sigma} - \tilde{R}_{\mu\nu\rho\sigma}\big) = \partial_\mu(\ldots)\,, \tag{6}$$

$\gamma_{\rm BI}$ can be traded for one of the parity-odd $\varepsilon \cdot T^2$ coefficients (d_{14} below).

3.2 Parity-Even Quadratic Sector (11 parameters)

3.2.1 Curvature: $\tilde{R}^2$ (6 terms)

$$\boxed{\mathcal{L}_{\tilde{R}^2} = \alpha_1\,\tilde{\mathcal{R}}^2 + \alpha_2\,\tilde{R}_{\mu\nu}\,\tilde{R}^{\mu\nu} + \alpha_3\,\tilde{R}_{\mu\nu}\,\tilde{R}^{\nu\mu} + \alpha_4\,\tilde{R}_{\mu\nu\rho\sigma}\,\tilde{R}^{\mu\nu\rho\sigma} + \alpha_5\,\tilde{R}_{\mu\nu\rho\sigma}\,\tilde{R}^{\mu\rho\nu\sigma} + \alpha_6\,\tilde{R}_{\mu\nu\rho\sigma}\,\tilde{R}^{\rho\sigma\mu\nu}} \tag{7}$$

where $\tilde{R}_{\mu\nu} \equiv \tilde{R}^{\lambda}{}_{\mu\lambda\nu}$ is the Ricci tensor (not symmetric in general). In the torsion-free limit, pair-exchange symmetry and the first Bianchi identity reduce these to 3 invariants, with the Gauss–Bonnet combination further reducing to 2. With torsion, all 6 are independent.

3.2.2 Torsion: T^2 (3 terms)

$$\boxed{\mathcal{L}_{T^2} = \beta_1\, T_{\lambda\mu\nu}\, T^{\lambda\mu\nu} + \beta_2\, T_{\lambda\mu\nu}\, T^{\mu\lambda\nu} + \beta_3\, T_\mu\, T^\mu} \tag{8}$$

In the irreducible basis (1): $\mathcal{L}_{T^2} = a\, q^2 + b\, S^2 + c\, T^2$, with $a = \beta_1 + \beta_2$, $\; b = -\frac{1}{6}(\beta_1 - \beta_2)$, $c = \beta_3 + \frac{1}{3}(2\beta_1 + \beta_2)$.

3.2.3 Electromagnetism: F^2 (1 term)

$$\boxed{\mathcal{L}_{F^2} = -\tfrac{1}{4}\, \gamma_1\, F_{\mu\nu}\, F^{\mu\nu}} \tag{9}$$

Setting $\gamma_1 = 1$ recovers standard Maxwell.

3.2.4 Curvature–EM cross-coupling: $\tilde{R} \times F$ (1 term)

$$\boxed{\mathcal{L}_{\tilde{R}F} = \delta_1\, \tilde{R}_{[\mu\nu]}\, F^{\mu\nu}} \tag{10}$$

This couples the antisymmetric Ricci tensor to $F^{\mu\nu}$. It vanishes identically when torsion is absent (where $\tilde{R}_{\mu\nu}$ is symmetric).

3.2.5 Vanishing cross-sectors

$\tilde{R} \times T$ (7 indices, odd) and $T \times F$ (5 indices, odd) yield no parity-even scalars.

3.3 Parity-Odd Quadratic Sector (21 terms; 20 physical)

Parity-odd invariants involve the Levi-Civita tensor $\varepsilon_{\mu\nu\rho\sigma}$. Barker et al. [1] omit them "only out of simplicity," noting "there are no convincing theoretical grounds for excluding them."

3.3.1 Pontryagin-type: $\varepsilon \cdot \tilde{R}^2$ (13 terms)

$$\begin{aligned}
\mathcal{L}_{\varepsilon\tilde{R}^2} = {} & d_1\, \varepsilon_{cdef}\, \tilde{R}_{ab}{}^{ef}\, \tilde{R}^{abcd} + d_2\, \varepsilon_{bdef}\, \tilde{R}_{ac}{}^{ef}\, \tilde{R}^{abcd} + d_3\, \varepsilon_{bdef}\, \tilde{R}_{a}{}^{e}{}_{c}{}^{f}\, \tilde{R}^{abcd} \\
& + d_4\, \varepsilon_{cdef}\, \tilde{R}^{ab}{}_{a}{}^{c}\, \tilde{R}_{b}{}^{def} + d_5\, \varepsilon_{abef}\, \tilde{R}^{abcd}\, \tilde{R}_{cd}{}^{ef} + d_6\, \varepsilon_{bdef}\, \tilde{R}^{ab}{}_{a}{}^{c}\, \tilde{R}_{c}{}^{def} \\
& + d_7\, \varepsilon_{bdef}\, \tilde{R}^{abcd}\, \tilde{R}_{c}{}^{e}{}_{a}{}^{f} + d_8\, \varepsilon_{abef}\, \tilde{R}^{abcd}\, \tilde{R}_{c}{}^{e}{}_{d}{}^{f} + d_9\, \varepsilon_{cdef}\, \tilde{R}^{ab}{}_{ab}\, \tilde{R}^{cdef} \\
& + d_{10}\, \varepsilon_{cdef}\, \tilde{R}^{ab}{}_{a}{}^{c}\, \tilde{R}^{de}{}_{b}{}^{f} + d_{11}\, \varepsilon_{bdef}\, \tilde{R}^{ab}{}_{a}{}^{c}\, \tilde{R}^{de}{}_{c}{}^{f} \\
& + d_{12}\, \varepsilon_{bcef}\, \tilde{R}^{ab}{}_{a}{}^{c}\, \tilde{R}^{de}{}_{d}{}^{f} + d_{13}\, \varepsilon_{abef}\, \tilde{R}^{abcd}\, \tilde{R}^{ef}{}_{cd}
\end{aligned} \tag{11}$$

The d_5 term is the generalized Pontryagin density. In the torsion-free limit, pair-exchange symmetry makes all 13 proportional to a single topological invariant. With torsion, they are generically independent and dynamical.

3.3.2 Nieh–Yan-type: $\varepsilon \cdot T^2$ (4 terms)

$$\boxed{\mathcal{L}_{\varepsilon T^2} = d_{14}\, \varepsilon^{\mu\nu\rho\sigma}\, T^{\alpha}{}_{\mu\nu}\, T_{\alpha\rho\sigma} + d_{15}\, \varepsilon^{\mu\nu\rho\sigma}\, T^{\alpha}{}_{\mu\rho}\, T_{\alpha\nu\sigma} + d_{16}\, \varepsilon^{\mu\nu\rho\sigma}\, T_\mu\, T_{\nu\rho\sigma} + d_{17}\, \varepsilon^{\mu\nu\rho\sigma}\, T^{\alpha}{}_{\mu\alpha}\, T_{\nu\rho\sigma}} \tag{12}$$

The d_{14} term is the Nieh–Yan form, related to the Holst term via (6).

3.3.3 EM theta term: $\varepsilon \cdot F^2$ (1 term, topological)

$$\mathcal{L}_{F\tilde{F}} = d_{18}\, \varepsilon^{\mu\nu\rho\sigma}\, F_{\mu\nu}\, F_{\rho\sigma} \tag{13}$$

This is $F \cdot \tilde{F}$, a total derivative. It does not contribute to field equations, reducing the physical count to **20**.

3.3.4 Parity-odd curvature–EM: $\varepsilon \cdot \tilde{R} \times F$ (3 terms)

$$\boxed{\mathcal{L}_{\varepsilon\tilde{R}F} = d_{19}\, \varepsilon^{\mu\nu\rho\sigma}\, \tilde{R}_{\mu\nu\rho}{}^{\alpha}\, F_{\sigma\alpha} + d_{20}\, \varepsilon^{\mu\nu\rho\sigma}\, \tilde{R}_{\mu\rho\nu}{}^{\alpha}\, F_{\sigma\alpha} + d_{21}\, \varepsilon^{\mu\nu\rho\sigma}\, \tilde{R}^{\alpha}{}_{\mu\alpha\nu}\, F_{\rho\sigma}} \tag{14}$$

All three vanish in the torsion-free limit.

3.4 Derivative Extensions

A covariant derivative ∇_α adds one index, enabling cross-sectors that vanish without it. Crucially, $\nabla T \sim \partial\Gamma \sim \tilde{R}$ (both $O(\partial^1\Gamma)$ in the first-order formalism), so terms built from $\{\tilde{R}, T, F, \nabla T\}$ are Yang–Mills type and ghost-free (Section 4.1).

3.4.1 Parity-even safe sectors

- $\nabla T \times F$ (3 terms, $\zeta_{1,2,3}$):

$$\mathcal{L}_{\nabla T\cdot F} = \zeta_1\, (\nabla T)^{ab}{}_{b}{}^{c}\, F_{ac} + \zeta_2\, (\nabla T)^{a}{}_{a}{}^{bc}\, F_{bc} + \zeta_3\, (\nabla T)^{ab}{}_{a}{}^{c}\, F_{bc} \tag{15}$$

  Torsion–EM cross-coupling, absent from both standard PGT and Barker's eWGT. Maps to Shapiro's $\theta_{2,5,6}$ [2].

- $\tilde{R} \times \nabla T$ ($\leq$ 10 terms†, $\chi_1 \ldots \chi_{10}$): All 10 contractions are listed in Appendix B. Barker's χ term [1] is the special case $\chi\, \tilde{R}_{[\mu\nu]}\, \partial^{[\mu} T^{\nu]}$.

- $(\nabla T)^2$ ($\leq$ 16 terms†, $\xi_1 \ldots \xi_{16}$): All 16 contractions are listed in Appendix B. Barker's ξ term ("Maxwell torsion") is the special case $-\frac{\xi}{12}\, \partial_{[\mu} T_{\nu]}\, \partial^{[\mu} T^{\nu]}$, the kinetic term enabling propagating vector torsion [1].

†Upper bounds before Bianchi reduction (Section 4.5).

3.4.2 Parity-odd safe sectors

- $\varepsilon \cdot \nabla T \times F$: 6 terms ($\tilde{\zeta}_{1\text{–}6}$)
- $\varepsilon \cdot \tilde{R} \times \nabla T$: $\leq$ 36 terms† ($\tilde{\chi}_{1\text{–}36}$)
- $\varepsilon \cdot (\nabla T)^2$: $\leq$ 38 terms† ($\tilde{\xi}_{1\text{–}38}$)

3.4.3 Ghost-prone sectors (excluded)

Terms involving $\nabla\tilde{R}$ or ∇F quadratically give 4th-order equations of motion (Ostrogradsky instability). These include $(\nabla F)^2$ (3), $\nabla F \times \nabla\tilde{R}$ (10), $(\nabla\tilde{R})^2$ (32), and their parity-odd counterparts. See Section 4.1 for the full classification.

3.5 Cubic Terms for Background Linearization

When linearizing around a background magnetic field $\bar{F}_{\mu\nu} \neq 0$, cubic Lagrangian terms contribute at quadratic order in perturbations: $\mathcal{L}_3(\epsilon x_1, \epsilon x_2, \bar{F}) = \epsilon^2\, x_1 \cdot x_2 \cdot \bar{B}_0$. These are independent coupling constants of the full theory, producing distinct position-dependent coefficients in the linearized PDE when $\bar{B}_0(\mathbf{x})$ varies in space (Section 5.2).

3.5.1 Parity-even cubic (safe, $n_\epsilon \leq 2$)

- $\tilde{R} \times F^2$ (4 terms, $\lambda_{1\text{–}4}$) — Gertsenshtein mechanism:

$$\begin{aligned}\mathcal{L}_{\tilde{R}FF} &= \lambda_1\, F^{ab} F^{cd} \tilde{R}_{abcd} + \lambda_2\, F^{ab} F^{cd} \tilde{R}_{acbd} \\ &\quad + \lambda_3\, F_a{}^c F^{ab} \tilde{R}_b{}^d{}_{cd} + \lambda_4\, F_{ab} F^{ab} \tilde{R}^{cd}{}_{cd}\end{aligned} \tag{16}$$

- $T^2 \times F$ (3 terms, $\kappa_{1\text{–}3}$) — torsion mass correction $\propto B_0$:

$$\mathcal{L}_{TTF} = \kappa_1\, F^{ab} T_a{}^{cd} T_{cbd} + \kappa_2\, F^{ab} T_{ab}{}^c T^d{}_{cd} + \kappa_3\, F^{ab} T^c{}_{ab} T^d{}_{cd} \tag{17}$$

- $\tilde{R}^2 \times F$ (8 terms, $\rho_{1\text{–}8}$): graviton self-interaction modified by B_0. All 8 listed in Appendix B.
- $\nabla T \times F^2$ (5 terms, $\mu_{1\text{–}5}$) — torsion–photon conversion $\propto B_0$:

$$\begin{aligned}\mathcal{L}_{\nabla T\cdot FF} &= \mu_1\, (\nabla T)^{abcd} F_{ac} F_{bd} + \mu_2\, (\nabla T)^{abcd} F_{ab} F_{cd} + \mu_3\, (\nabla T)^{ab}{}_b{}^c F_a{}^d F_{cd} \\ &\quad + \mu_4\, (\nabla T)^{ab}{}_a{}^c F_b{}^d F_{cd} + \mu_5\, (\nabla T)^{ab}{}_{ab} F_{cd} F^{cd}\end{aligned} \tag{18}$$

3.5.2 Parity-odd cubic (safe, $n_\epsilon \leq 2$)

- $\varepsilon \cdot \tilde{R} \times F^2$: 12 terms ($\tilde{\lambda}_{1\text{–}12}$)
- $\varepsilon \cdot T^2 \times F$: 15 terms ($\tilde{\kappa}_{1\text{–}15}$); maps to Shapiro's $\theta_{1,3}$ [2]
- $\varepsilon \cdot F^2 \times \nabla T$: 18 terms ($\tilde{\mu}_{1\text{–}18}$)

3.5.3 Mixing classification

3.6 Chern–Simons Torsion–EM Coupling (3 parameters)

There is one gauge-invariant structure involving bare A_μ: $\mathcal{L}_{\mathrm{CS}} = \varepsilon^{\mu\nu\rho\sigma}\, V_\mu\, A_\nu\, F_{\rho\sigma}$, where V_μ is a torsion-derived vector. Contracting with the three irreducible torsion vectors gives:

$$\mathcal{L}_{\mathrm{CS}}^{(1)} = \chi_1^{\mathrm{CS}}\, \varepsilon^{\mu\nu\rho\sigma}\, T_\mu\, A_\nu\, F_{\rho\sigma} \qquad \text{(trace torsion)} \tag{19}$$

$$\mathcal{L}_{\mathrm{CS}}^{(2)} = \chi_2^{\mathrm{CS}}\, \varepsilon^{\mu\nu\rho\sigma}\, S_\mu\, A_\nu\, F_{\rho\sigma} \qquad \text{(axial torsion)} \tag{20}$$

$$\mathcal{L}_{\mathrm{CS}}^{(3)} = \chi_3^{\mathrm{CS}}\, \varepsilon^{\mu\nu\rho\sigma}\, q^\lambda{}_{\mu\alpha}\, g^\alpha{}_\nu\, A_\lambda\, F_{\rho\sigma} \qquad \text{(tensor torsion)} \tag{21}$$

The axial coupling χ_2^{CS} is derived by Obukhov et al. [3] from the minimal coupling prescription; the torsion sources the magnetic helicity density $\mathbf{A} \cdot \mathbf{B}$.

Table 1: Cubic sectors classified by mixing type when linearized around $\bar{B}_0 \neq 0$, $\bar{T} = 0$.

Sector	**Count**	**Mixing type**	**Mechanism**
$\tilde{R} \times F^2$	4	$h \leftrightarrow f$	One $F \to \bar{F}$
$\tilde{R}^2 \times F$	8	h self-int. mod B_0	One $F \to \bar{F}$
$T^2 \times F$	3	t mass corr. $\propto B_0$	Both $T \to \epsilon t$
$\nabla T \times F^2$	5	$t \leftrightarrow f$	One $F \to \bar{F}$

Scope of the enumeration. This catalog covers terms up to **cubic field-strength count**, following the standard PGT convention of restricting to actions at most quadratic in field strengths [?, ?]. All cubic terms listed here are ghost-safe (they produce $\leq$2nd-order equations of motion). Note, however, that the ghost-safety criterion alone does *not* give a finite set of allowed operators: terms with higher powers of T (e.g. T^4, T^6) also produce $\leq$2nd-order equations because T carries no additional derivatives of the connection. These higher-T terms are excluded by convention, not by ghost safety. Whether they are physically relevant is an open question.

Note on EFT ordering. A proper effective field theory classification would assign mass dimensions to the fields and organise operators by their dimension. However, assigning mass dimensions requires first solving the propagating mass spectrum of the specific theory (Stückelberg decomposition, propagating mode identification), which depends on the theory parameters [?]. EFT ordering is deferred to future work. See `docs/tex/background_validity.tex`, §??.

3.7 Master Summary

4 Constraints on the Parameter Space

4.1 Ghost Freedom and Derivative Counting

In the first-order (Palatini) formalism, Γ is an independent dynamical variable. The derivative orders are:

$$\tilde{R} \sim O(\partial^1\Gamma)\,, \qquad T \sim O(\partial^0\Gamma)\,, \qquad F \sim O(\partial^1 A)\,, \qquad \nabla T \sim O(\partial^1\Gamma)\,. \tag{22}$$

Since $\nabla T \sim \partial\Gamma \sim \tilde{R}$, both are first-order in the gauge field. Products of $\{\tilde{R}, T, F, \nabla T\}$ give at most second-order equations of motion: **Yang–Mills type**, safe from Ostrogradsky ghosts.

In contrast, $\nabla\tilde{R} \sim \partial^2\Gamma$ and $\nabla F \sim \partial^2 A$ are higher-derivative. Any term quadratic in $\nabla\tilde{R}$ or ∇F gives fourth-order equations $\Rightarrow$ Ostrogradsky instability. The full enumeration finds **31 safe sectors** (975 contractions), **11 borderline** (1,048), and **15 ghost-prone** (3,589).

4.2 Propagator Constraints

Barker's Master Constraint. Even within the safe $\tilde{R}^2$ sector, the propagator contains a quartic pole (k^4) unless [1]:

$$(2\alpha_2 + 4\alpha_4 + \alpha_5)\,\xi = \chi^2\,. \tag{23}$$

Table 2: Complete parameter count for the general PGT+EM Lagrangian. "Status": safe = ghost-free (Yang–Mills type); topo = topological (total derivative). Counts marked † are upper bounds before Bianchi reduction. "Lit." shows equivalents in Barker (B), Shapiro (S), Obukhov (O), Hayashi–Shirafuji (HS).

Sector	**Symbols**	**Count**	**Status**	**Lit.**
Linear (3 parameters)				
Einstein–Hilbert	$M_{\rm Pl}$	1	fund.	all
Cosmological constant	Λ	1	fund.	all
Holst	$\gamma_{\rm BI}$	1	fund.	—
Parity-even quadratic, non-derivative (11 parameters)				
$\tilde{R}^2$	$\alpha_1 \ldots \alpha_6$	6	safe	B: $\alpha_{1\text{–}6}$
T^2	$\beta_1, \beta_2, \beta_3$	3	safe	B: $\beta_{1\text{–}3}$; S: $I_{1\text{–}3}$
F^2	γ_1	1	safe	—
$\tilde{R} \times F$	δ_1	1	safe	—
Parity-odd quadratic, non-derivative (21; 20 physical)				
$\varepsilon \cdot \tilde{R}^2$	$d_1 \ldots d_{13}$	13	safe	—
$\varepsilon \cdot T^2$	$d_{14} \ldots d_{17}$	4	safe	—
$\varepsilon \cdot F^2$	d_{18}	1	topo	—
$\varepsilon \cdot \tilde{R} \times F$	d_{19}, d_{20}, d_{21}	3	safe	—
Derivative extensions, parity-even (safe)				
$\nabla T \times F$	$\zeta_1, \zeta_2, \zeta_3$	3	safe	S: $\theta_{2,5,6}$
$\tilde{R} \times \nabla T$	$\chi, \chi_2 \cdots \chi_{10}$	$10^\dagger$	safe	B: χ
$(\nabla T)^2$	$\xi, \xi_2 \ldots \xi_{16}$	$16^\dagger$	safe	B: ξ
Derivative extensions, parity-odd (safe)				
$\varepsilon \cdot \nabla T \times F$	$\tilde{\zeta}_{1\text{–}6}$	6	safe	—
$\varepsilon \cdot \tilde{R} \times \nabla T$	$\tilde{\chi}_{1\text{–}36}$	$36^\dagger$	safe	—
$\varepsilon \cdot (\nabla T)^2$	$\tilde{\xi}_{1\text{–}38}$	$38^\dagger$	safe	—
Cubic, safe, $O(\epsilon^2)$-relevant				
$\tilde{R} \times F^2$	$\lambda_{1\text{–}4}$	4	safe	—
$T^2 \times F$	$\kappa_{1\text{–}3}$	3	safe	S: θ_4
$\tilde{R}^2 \times F$	$\rho_{1\text{–}8}$	8	safe	—
$\nabla T \times F^2$	$\mu_{1\text{–}5}$	5	safe	—
$\varepsilon \cdot \tilde{R} \times F^2$	$\tilde{\lambda}_{1\text{–}12}$	12	safe	—
$\varepsilon \cdot T^2 \times F$	$\tilde{\kappa}_{1\text{–}15}$	15	safe	S: $\theta_{1,3}$
$\varepsilon \cdot F^2 \times \nabla T$	$\tilde{\mu}_{1\text{–}18}$	18	safe	—
Chern–Simons (gauge potential)				
$\varepsilon \cdot T \cdot A \cdot F$	$\chi_1^{\rm CS}$	1	safe	—
$\varepsilon \cdot S \cdot A \cdot F$	$\chi_2^{\rm CS}$	1	safe	O: χ
$\varepsilon \cdot q \cdot A \cdot F$	$\chi_3^{\rm CS}$	1	safe	—

Shapiro's unitarity condition. For propagating torsion, longitudinal and transverse modes cannot propagate simultaneously without a ghost [2]. The coefficient of $(\partial_\mu S^\mu)^2$ must vanish ($b = 0$), leaving only transverse (Maxwell-like) torsion propagation.

Blagojević–Cvetković ghost-free catalog. Blagojević & Cvetković [5] provide a systematic computer-aided classification of the full 10-parameter parity-preserving PGT (parameters $\lambda, r_{1\text{–}5}, t_{1\text{–}3}$ in their notation, mapping to our $M_{\rm Pl}, \alpha_{1\text{–}6}, \beta_{1\text{–}3}$). Of 1918 critical parameter cases (submanifolds with enhanced gauge symmetry), **450 are ghost-and-tachyon free**, including 10 that are power-counting renormalizable. Key results for sweep constraints:

- **Propagating 0^- (pseudoscalar) tordion:** ghost-free with $r_2 < 0$, $t_2 > 0$, $\lambda > 0$; requires Riemann-squared coupling ($r_2 R^{ABCD} R_{ABCD}$ type, mapping to α_4), *not* $\tilde{\mathcal{R}}^2$. Mass: $m^2_{0^-} = -t_2/r_2$.
- **Propagating 0^+ (scalar) tordion:** ghost-free in critical cases e.g. $r_1 = 0$, $2r_3 = r_4 = -r_5$, with $r_3 > 0$, $r_2 < 0$, $t_2 > 0$, $t_3(\lambda - t_3) < 0$.
- **Propagating 0^- without $\tilde{\mathcal{R}}^2$:** all curvature-squared couplings except r_2 can be zero. This avoids the Ostrogradsky ghost from $b_5 \widetilde{R}^2$ entirely.

Nikiforova et al. [10] provide complementary stability conditions around Minkowski: $c_1 > 0$, $c_5 < 0$, $c_6 > 0$, $\alpha < 0$, $\tilde{\alpha} > 0$, with standard massive Klein–Gordon dispersion for both spin-0 and spin-2 torsion modes.

Linearised regime and ghost onset. Aoki [11] shows that PGT is equivalent to ghost-free massive bigravity at the linearised level; ghosts enter only at cubic order in perturbations, with cutoff $\Lambda_4 = (M_{\rm Pl} m^3)^{1/4}$. For TIDAL's linearised simulations, the quadratic PGT is therefore **healthy** even with R^2 terms — the Ostrogradsky ghost is a *non-linear* artefact.

These results should be encoded as constraint sets in `tidal sweep` to restrict parameter scans to ghost-free windows (see Priority Parameters, §**??**).

4.3 Topological Terms

With torsion, the Gauss–Bonnet combination is **not topological** (pair-exchange symmetry fails), so all 6 $\tilde{R}^2$ parameters survive. The one topological term is $\varepsilon \cdot F^2$ ($F \cdot \tilde{F}$), reducing the parity-odd count: $13 + 4 + 3 = 20$ physical (not 21).

4.4 Dimensional-Dependent Identities (DDIs)

In $D = 4$ dimensions, antisymmetrization over $D+1 = 5$ or more indices of the metric vanishes identically. This generates *additional* linear relations among tensor contractions that hold only in specific dimensions, beyond the tensor symmetries used by `AllContractions`. The `xTras` function `ConstructDDIs` identifies these.

The core parity-even sectors ($\widetilde{R}^2$, T^2, F^2, $\tilde{R} \times F$) have ≤ 8 pre-contraction indices and are **not affected** by DDIs in 4D — their counts (6, 3, 1, 1) are exact, as confirmed both by `ConstructDDIs` (returning 0 identities) and by the standard PGT literature [1, 6].

The parity-odd sectors involving ε have 10–12 pre-contraction indices and **are subject to DDIs**: `ConstructDDIs` returns non-trivial identities for $\varepsilon \cdot \tilde{R}^2$, $\varepsilon \cdot T^2$, $\varepsilon \cdot \tilde{R} \times F$, and all parity-odd derivative sectors. The raw DDI counts are large (often exceeding the contraction counts), indicating extensive linear dependencies among the DDIs themselves. The *actual reduction* per sector requires computing the rank of the DDI system when expressed in the basis of `AllContractions` — an open computation.

Consequently, our parity-odd counts (**13, 4, 3** for the core; **6, 36, 38** for derivatives) are confirmed as **upper bounds** that may decrease once DDIs are fully resolved. The parity-even counts (**6, 3, 1, 1**) are **exact**.

4.5 Bianchi Identities

The first Bianchi identity in Riemann–Cartan geometry,

$$\tilde{R}^{\lambda}{}_{[\mu\nu\rho]} = \nabla_{[\mu} T^{\lambda}{}_{\nu\rho]} + T^{\lambda}{}_{\sigma[\mu} T^{\sigma}{}_{\nu\rho]} \,, \tag{24}$$

provides linear relations among $\tilde{R}^2$, $\tilde{R}\times\nabla T$, and $(\nabla T)^2$ scalar invariants that `AllContractions` does not account for. Thus, the derivative sector counts $(10 + 16 + 6 = 32)$ are **upper bounds**. Barker's χ and ξ are confirmed independent [1]; the exact reduced count for the remaining terms is an open computation.

4.6 IBP Relations and Field Redefinitions

Derivative sectors come in IBP pairs (e.g., $\nabla T \times F$ and $T \times \nabla F$). In the ghost-free theory, only the member built from safe building blocks survives (e.g., $\nabla T \times F$ rather than $T \times \nabla F$). The classification identifies **10 IBP-redundant sectors**.

4.7 Projective Invariance (Optional)

The connection $\Gamma^{\lambda}{}_{\mu\nu}$ has a projective freedom: $\Gamma^{\lambda}{}_{\mu\nu} \to \Gamma^{\lambda}{}_{\mu\nu} + \delta^{\lambda}_{\mu}\,\xi_{\nu}$ for arbitrary ξ_{ν}. Under this, the torsion trace shifts: $T_{\mu} \to T_{\mu} + (D{-}1)\,\xi_{\mu}$, while the axial ($S_{\mu}$) and tensor ($q_{\alpha\beta\gamma}$) parts are invariant.

If the action is required to be **projective-invariant**, then the torsion trace T_{μ} is pure gauge and must decouple. In the irreducible basis, this means the coefficient c of $T_{\mu}T^{\mu}$ must vanish:

$$c = \beta_3 + \tfrac{1}{3}(2\beta_1 + \beta_2) = 0 \,, \tag{25}$$

reducing the torsion sector from 3 to **2 independent parameters** (tensor a and axial b only). Similarly, all derivative and cubic sectors involving the torsion trace would be constrained.

Projective invariance is *not* a universal requirement — it is a symmetry choice. Many physically interesting theories (including Barker's vector torsion propagation via ξ) break projective invariance explicitly. We include it as an *optional* constraint for theories where only tensor and axial torsion are desired.

Shapiro [2] shows that $A_{\mu} \to A_{\mu} + \eta_2 T_{\mu}$ absorbs one torsion–EM coupling parameter.

5 Application: Linearization Around a Magnetic Background

We linearize around flat spacetime with $\bar{B}_0 \neq 0$ and $\bar{T} = 0$ (Gertsenshtein-type setting).

5.1 How Cubic Terms Contribute

A cubic term $\mathcal{L}_3(X, Y, F)$ with $F = \bar{F} + \epsilon f$ contributes at $O(\epsilon^2)$:

- $\mathcal{L}_3(\epsilon h,\, \epsilon f,\, \bar{F})$: graviton–photon mixing $\propto B_0$
- $\mathcal{L}_3(\epsilon t,\, \epsilon f,\, \bar{F})$: torsion–photon mixing $\propto B_0$
- $\mathcal{L}_3(\epsilon h,\, \epsilon t,\, \bar{F})$: graviton–torsion mixing $\propto B_0$

Quadratic terms like $\nabla T \times F$ also produce mixing directly (no background needed for the $O(\epsilon^2)$ piece $\nabla t \cdot f$).

5.2 Independence of Cubic Parameters

Cubic Lagrangian parameters (λ_I, κ_I, etc.) are **independent coupling constants** of the full theory:

1. **Different operators:** The 4 terms in $\tilde{R} \times F^2$ couple the *full 4-index Riemann* to $\bar{F} \cdot f$, while $\tilde{R} \times F$ couples only the antisymmetric Ricci.
2. **Spatially varying $\bar{B}_0(\mathbf{x})$:** Each contraction generates a different position-dependent coefficient in the linearized PDE, plus gradient terms $\partial_\mu \bar{F}_{\alpha\beta}$ with no quadratic analog.
3. **Parameter sweeps:** Absorbing cubics into effective quadratics would destroy independent exploration of their physical effects.

5.3 Perturbation-Order Filtering

Each $\tilde{R}$ or T factor contributes $O(\epsilon)$; each F contributes $O(1)$ (background) or $O(\epsilon)$ (perturbation), with at most one background. The perturbation order is $n_\epsilon = n_{\tilde{R}} + n_T + \max(0, n_F - 1)$. For $O(\epsilon^2)$: $n_\epsilon \leq 2$, which eliminates most cubic sectors. Only $\sim$200 of the 5,612 total contractions survive.

5.4 Metric Expansion and the Gertsenshtein Mechanism

Every Lagrangian term implicitly couples to $h_{\mu\nu}$ through the metric expansion $g^{\mu\nu} = \eta^{\mu\nu} - \epsilon h^{\mu\nu} + O(\epsilon^2)$. The Maxwell term F^2, expanded, generates the cubic coupling $h \cdot F \cdot F$ that produces graviton–photon mixing $\propto B_0$: the **Gertsenshtein mechanism**. This is automatic in the TIDAL pipeline (`xPert` expansion with `[[background_fields]]`).

6 Literature Cross-References

6.1 Barker's eWGT Completion

The standard PGT action contains only $\mathcal{L}_{\tilde{R}^2} + \mathcal{L}_{T^2}$ (9 parameters). Barker et al. [1] showed this is *incomplete*: the scale-invariant (eWGT) embedding generates two additional terms,

$$\chi \, \tilde{R}_{[\mu\nu]} \, \partial^{[\mu} T^{\nu]} - \frac{\xi}{12} \, \partial_{[\mu} T_{\nu]} \, \partial^{[\mu} T^{\nu]} \,, \tag{26}$$

which are special cases of our $\tilde{R} \times \nabla T$ and $(\nabla T)^2$ sectors. The ξ term enables propagating vector torsion, resolving a 25-year no-go result.

6.2 Shapiro's Non-Minimal EM–Torsion Couplings

Shapiro [2] classifies all gauge-invariant non-minimal EM–torsion couplings as $\mathcal{L}_{\rm nm} = F_{\mu\nu}K^{\mu\nu}$ with 6 parameters $\theta_{1\text{–}6}$. These map to our sectors: $\theta_{2,5,6} \to \nabla T \times F$ (parity-even), $\theta_{1,3} \to \varepsilon \cdot T^2 \times F$ (parity-odd cubic), $\theta_4 \to T^2 \times F$ (parity-even cubic).

7 Discussion and Open Questions

7.1 Hierarchy of Generality

Table 3: Nested parameter spaces, from minimal to maximal. †Upper bounds before Bianchi reduction.

Theory	**Parameters**	**Content**
Einstein–Cartan	3	$M_{\rm Pl}, \Lambda, \gamma_{\rm BI}$
Standard PGT (parity-even)	12	$+\alpha_{1\text{–}6}, \beta_{1\text{–}3}$
PGT + EM coupling	14	$+\gamma_1, \delta_1$
+ parity-odd (physical)	34	$+d_{1\text{–}13}, d_{14\text{–}17}, d_{19\text{–}21}$
+ Barker eWGT	36	$+\chi, \xi$
+ all safe ∇T sectors†	$\lesssim 67$	$+\zeta, \chi_{2\text{–}10}, \xi_{2\text{–}16}, \tilde{\zeta}, \tilde{\chi}, \tilde{\xi}$
+ safe cubics ($n_\epsilon \leq 2$)	+65	$\lambda, \kappa, \rho, \mu$ + parity-odd
+ Chern–Simons	+3	$\chi^{\rm CS}_{1,2,3}$

7.2 Priority Parameters for Sweeps

For initial parameter sweeps with magnetic background $\bar{B}_0$:

1. $\beta_{1,2,3}$: simplest torsion dynamics (3 params)
2. χ, ξ: Barker's propagating vector torsion (2 params)
3. $\zeta_{1,2,3}$: torsion–EM cross-coupling (3 params)
4. $\alpha_{1\ldots6}$: graviton–torsion mixing (6 params)
5. $\lambda_{1\ldots4}$: Gertsenshtein cubic (4 params)
6. $\kappa_{1,2,3}$: torsion mass correction $\propto B_0$ (3 params)
7. $d_{14\ldots17}$: CP-violating torsion (4 params)
8. $\chi_2^{\rm CS}$: axial torsion × magnetic helicity (1 param)

7.3 Open Questions

1. **Bianchi reduction:** How many of the $10 + 16 = 26$ safe derivative terms survive the first Bianchi identity (24)? Barker's χ, ξ survive; the rest is open.

2. **Parity-odd propagator analysis:** Ghost/tachyon-free windows for the 20 parity-odd parameters require propagator analysis (PSALTer). Partial results exist [4, 5]. For the parity-even sector, Blagojević & Cvetković's 450 ghost-free cases (§4.2) provide concrete constraint sets for parameter sweeps.

3. **Nieh–Yan/Holst interplay:** d_{14} and γ_{BI} are not simultaneously independent. Choose one as fundamental.

4. **Notation finalization:** Confirm the d_I scheme for parity-odd terms and the χ_I/ξ_I extension of Barker's notation.

5. **Quartic terms:** Terms $O(\text{field}^4)$ could contribute at $O(\epsilon^2)$ when two factors take background values. Beyond current scope for $\bar{T} = 0$ (quartic F^2T^2 terms vanish identically). If background torsion $\bar{T} \neq 0$ is considered, the 17 independent F^2T^2 structures classified by Itin [12] become relevant. Separately, Bahamonde et al. [13] show that cubic $R{\cdot}T{\cdot}T$ invariants (26 terms, 6 constrained) eliminate ghost pathologies in the axial and vector torsion sectors around arbitrary (non-flat) backgrounds. These are beyond the linearised flat-background regime but relevant if TIDAL extends to curved-background simulations.

A Algorithmic Enumeration via `xTras`

The enumeration uses `AllContractions[expr]` from the `xTras` package [9] for `xAct`. Given a tensorial expression with free indices:

1. Extract all free indices and their tensor symmetries.

2. Construct the strong generating set (SGS) of the symmetry group.

3. Call the external `xPerm` executable to enumerate all valid index permutations modulo the symmetry group.

4. For each permutation, reconstruct the tensorial expression with metric contractions for paired indices.

5. Canonicalize via `ToCanonical` to remove algebraic equivalences.

6. Return the list of unique, independent contractions.

The wrapper `MakeContractionAnsatz[expr]` assigns constant coefficients to each contraction.

An automated loop generates all unordered pairs (quadratic) and triples (cubic) from $\{\tilde{R}, T, F, \nabla\tilde{R}, \nabla T, \nabla F\}$, checks index parity (even $\Rightarrow$ contractible), and calls `AllContractions` on each. This yields **57 non-zero sectors** with **5,612 total** independent contractions. The full sector-by-sector results are in `enumeration_classified.json`.

B Complete Explicit Contractions

All contractions listed below are produced by `AllContractions` from `xTras` and represent the canonical forms in 4D. The raw Mathematica output is in `explicit_terms_raw.txt`.

Index notation. The torsion tensor always carries three indices $T^\lambda{}_{\mu\nu}$. Where a pair is contracted (e.g., $T^d{}_{cd}$), this denotes the *torsion trace* $T_c \equiv T^\lambda{}_{\lambda c}$. Likewise, $(\nabla T)^{ab}{}_a{}^c$ denotes the trace $(\nabla_d T^{dbc}) = \nabla_d T^{dbc}$ (with one index of the derivative–torsion pair contracted). Where two such contractions appear in a single term (e.g., $T^d{}_{cd}\, T^e{}_{ae}$), both factors are torsion traces $T_c\, T_a$.

B.1 $\tilde{R} \times \nabla T$ (10 terms)

$$\begin{array}{ll}
\chi_1\, (\nabla T)^{abcd}\, \tilde{R}_{abcd} & \chi_2\, (\nabla T)^{abcd}\, \tilde{R}_{acbd} \\
\chi_3\, (\nabla T)^{ab}{}_b{}^c\, \tilde{R}_a{}^d{}_{cd} & \chi_4\, (\nabla T)^{abcd}\, \tilde{R}_{bcad} \\
\chi_5\, (\nabla T)^a{}_a{}^{bc}\, \tilde{R}_b{}^d{}_{cd} & \chi_6\, (\nabla T)^{ab}{}_a{}^c\, \tilde{R}_b{}^d{}_{cd} \\
\chi_7\, (\nabla T)^{abcd}\, \tilde{R}_{cdab} & \chi_8\, (\nabla T)^{ab}{}_b{}^c\, \tilde{R}_c{}^d{}_{ad} \\
\chi_9\, (\nabla T)^{ab}{}_a{}^c\, \tilde{R}_c{}^d{}_{bd} & \chi_{10}\, (\nabla T)^{ab}{}_{ab}\, \tilde{R}^{cd}{}_{cd}
\end{array} \tag{27}$$

Barker's χ term. $\chi\, \tilde{R}_{[\mu\nu]}\, \partial^{[\mu} T^{\nu]}$ involves only the antisymmetric Ricci $\tilde{R}_{[\mu\nu]} = \tilde{R}^\lambda{}_{[\mu|\lambda|\nu]}$ and the torsion-trace field strength $\partial_{[\mu} T_{\nu]}$. In our canonical basis, these are the *trace-involving* terms: χ_5 ($(\nabla T)^a{}_a{}^{bc}\, \tilde{R}_b{}^d{}_{cd}$), χ_6 ($(\nabla T)^{ab}{}_a{}^c\, \tilde{R}_b{}^d{}_{cd}$), χ_9 ($(\nabla T)^{ab}{}_a{}^c\, \tilde{R}_c{}^d{}_{bd}$), and χ_{10} ($(\nabla T)^{ab}{}_{ab}\, \tilde{R}^{cd}{}_{cd}$). Barker's χ is a specific linear combination of these four.

B.2 $(\nabla T)^2$ (16 terms)

$$\begin{array}{ll}
\xi_1\, (\nabla T)_a{}^d{}_{cd}\, (\nabla T)^{ab}{}_b{}^c & \xi_2\, (\nabla T)_{abcd}\, (\nabla T)^{abcd} \\
\xi_3\, (\nabla T)_{acbd}\, (\nabla T)^{abcd} & \xi_4\, (\nabla T)^{abcd}\, (\nabla T)_{bacd} \\
\xi_5\, (\nabla T)^{abcd}\, (\nabla T)_{bcad} & \xi_6\, (\nabla T)^a{}_a{}^{bc}\, (\nabla T)_b{}^d{}_{cd} \\
\xi_7\, (\nabla T)^{ab}{}_a{}^c\, (\nabla T)_b{}^d{}_{cd} & \xi_8\, (\nabla T)^{abcd}\, (\nabla T)_{cbad} \\
\xi_9\, (\nabla T)^{abcd}\, (\nabla T)_{cdab} & \xi_{10}\, (\nabla T)^{ab}{}_b{}^c\, (\nabla T)_c{}^d{}_{ad} \\
\xi_{11}\, (\nabla T)^{ab}{}_a{}^c\, (\nabla T)_c{}^d{}_{bd} & \xi_{12}\, (\nabla T)^{ab}{}_{ab}\, (\nabla T)^{cd}{}_{cd} \\
\xi_{13}\, (\nabla T)^a{}_a{}^{bc}\, (\nabla T)^d{}_{bcd} & \xi_{14}\, (\nabla T)^{ab}{}_a{}^c\, (\nabla T)^d{}_{bcd} \\
\xi_{15}\, (\nabla T)^{ab}{}_a{}^c\, (\nabla T)^d{}_{cbd} & \xi_{16}\, (\nabla T)^a{}_a{}^{bc}\, (\nabla T)^d{}_{dbc}
\end{array} \tag{28}$$

Barker's ξ term. $-\frac{\xi}{12}\, \partial_{[\mu} T_{\nu]}\, \partial^{[\mu} T^{\nu]}$ involves only the torsion-trace field strength $H_{\mu\nu} = 2\partial_{[\mu}(T_{\nu]}/3)$. In our canonical basis, these are the *double-trace* terms where both (∇T) factors have a contracted pair: ξ_6, ξ_7, ξ_{12}, and ξ_{16}. Barker's ξ is a specific linear combination of these four.

B.3 $\tilde{R}^2 \times F$ (8 terms)

$$\begin{array}{ll}
\rho_1\, F^{ab}\, \tilde{R}_a{}^{cde}\, \tilde{R}_{cdbe} & \rho_2\, F^{ab}\, \tilde{R}_{ab}{}^{cd}\, \tilde{R}_c{}^e{}_{de} \\
\rho_3\, F^{ab}\, \tilde{R}_a{}^c{}_b{}^d\, \tilde{R}_c{}^e{}_{de} & \rho_4\, F^{ab}\, \tilde{R}_c{}^e{}_{de}\, \tilde{R}^{cd}{}_{ab} \\
\rho_5\, F^{ab}\, \tilde{R}_a{}^{cde}\, \tilde{R}_{debc} & \rho_6\, F^{ab}\, \tilde{R}_a{}^c{}_c{}^d\, \tilde{R}_d{}^e{}_{be} \\
\rho_7\, F^{ab}\, \tilde{R}_a{}^c{}_b{}^d\, \tilde{R}_d{}^e{}_{ce} & \rho_8\, F^{ab}\, \tilde{R}_a{}^c{}_{bc}\, \tilde{R}^{de}{}_{de}
\end{array} \tag{29}$$

B.4 $\varepsilon \cdot \nabla T \times F$ (6 terms)

$$\begin{array}{ll}
\tilde{\zeta}_1\,(\nabla T)^{abcd}\,\varepsilon_{bcde}\,F_a{}^e & \tilde{\zeta}_2\,(\nabla T)^{abcd}\,\varepsilon_{acde}\,F_b{}^e \\
\tilde{\zeta}_3\,(\nabla T)^{abcd}\,\varepsilon_{abde}\,F_c{}^e & \tilde{\zeta}_4\,(\nabla T)^{ab}{}_b{}^c\,\varepsilon_{acde}\,F^{de} \\
\tilde{\zeta}_5\,(\nabla T)^a{}_a{}^{bc}\,\varepsilon_{bcde}\,F^{de} & \tilde{\zeta}_6\,(\nabla T)^{ab}{}_a{}^c\,\varepsilon_{bcde}\,F^{de}
\end{array} \qquad (30)$$

B.5 $\varepsilon \cdot \tilde{R} \times \nabla T$ (36 terms)

$$\begin{array}{ll}
\tilde{\chi}_1\,(\nabla T)^{abcd}\,\varepsilon_{cdef}\,\tilde{R}_{ab}{}^{ef} & \tilde{\chi}_2\,(\nabla T)^{abcd}\,\varepsilon_{bdef}\,\tilde{R}_{ac}{}^{ef} \\
\tilde{\chi}_3\,(\nabla T)^{ab}{}_b{}^c\,\varepsilon_{cdef}\,\tilde{R}_a{}^{def} & \tilde{\chi}_4\,(\nabla T)^{abcd}\,\varepsilon_{cdef}\,\tilde{R}_a{}^e{}_b{}^f \\
\tilde{\chi}_5\,(\nabla T)^{abcd}\,\varepsilon_{bdef}\,\tilde{R}_a{}^e{}_c{}^f & \tilde{\chi}_6\,(\nabla T)^{abcd}\,\varepsilon_{bcdf}\,\tilde{R}_a{}^e{}_e{}^f \\
\tilde{\chi}_7\,(\nabla T)^{abcd}\,\varepsilon_{adef}\,\tilde{R}_{bc}{}^{ef} & \tilde{\chi}_8\,(\nabla T)^a{}_a{}^{bc}\,\varepsilon_{cdef}\,\tilde{R}_b{}^{def} \\
\tilde{\chi}_9\,(\nabla T)^{ab}{}_a{}^c\,\varepsilon_{cdef}\,\tilde{R}_b{}^{def} & \tilde{\chi}_{10}\,(\nabla T)^{abcd}\,\varepsilon_{cdef}\,\tilde{R}_b{}^e{}_a{}^f \\
\tilde{\chi}_{11}\,(\nabla T)^{abcd}\,\varepsilon_{adef}\,\tilde{R}_b{}^e{}_c{}^f & \tilde{\chi}_{12}\,(\nabla T)^{abcd}\,\varepsilon_{acdf}\,\tilde{R}_b{}^e{}_e{}^f \\
\tilde{\chi}_{13}\,(\nabla T)^{abcd}\,\varepsilon_{abef}\,\tilde{R}_{cd}{}^{ef} & \tilde{\chi}_{14}\,(\nabla T)^{ab}{}_b{}^c\,\varepsilon_{adef}\,\tilde{R}_c{}^{def} \\
\tilde{\chi}_{15}\,(\nabla T)^{ab}{}_a{}^c\,\varepsilon_{bdef}\,\tilde{R}_c{}^{def} & \tilde{\chi}_{16}\,(\nabla T)^{abcd}\,\varepsilon_{bdef}\,\tilde{R}_c{}^e{}_a{}^f \\
\tilde{\chi}_{17}\,(\nabla T)^{abcd}\,\varepsilon_{adef}\,\tilde{R}_c{}^e{}_b{}^f & \tilde{\chi}_{18}\,(\nabla T)^{abcd}\,\varepsilon_{abef}\,\tilde{R}_c{}^e{}_d{}^f \\
\tilde{\chi}_{19}\,(\nabla T)^{abcd}\,\varepsilon_{abdf}\,\tilde{R}_c{}^e{}_e{}^f & \tilde{\chi}_{20}\,(\nabla T)^{ab}{}_{ab}\,\varepsilon_{cdef}\,\tilde{R}^{cdef} \\
\tilde{\chi}_{21}\,(\nabla T)^{ab}{}_b{}^c\,\varepsilon_{cdef}\,\tilde{R}^{de}{}_a{}^f & \tilde{\chi}_{22}\,(\nabla T)^a{}_a{}^{bc}\,\varepsilon_{cdef}\,\tilde{R}^{de}{}_b{}^f \\
\tilde{\chi}_{23}\,(\nabla T)^{ab}{}_a{}^c\,\varepsilon_{cdef}\,\tilde{R}^{de}{}_b{}^f & \tilde{\chi}_{24}\,(\nabla T)^{ab}{}_b{}^c\,\varepsilon_{adef}\,\tilde{R}^{de}{}_c{}^f \\
\tilde{\chi}_{25}\,(\nabla T)^{ab}{}_a{}^c\,\varepsilon_{bdef}\,\tilde{R}^{de}{}_c{}^f & \tilde{\chi}_{26}\,(\nabla T)^{ab}{}_b{}^c\,\varepsilon_{acef}\,\tilde{R}^{de}{}_d{}^f \\
\tilde{\chi}_{27}\,(\nabla T)^a{}_a{}^{bc}\,\varepsilon_{bcef}\,\tilde{R}^{de}{}_d{}^f & \tilde{\chi}_{28}\,(\nabla T)^{ab}{}_a{}^c\,\varepsilon_{bcef}\,\tilde{R}^{de}{}_d{}^f \\
\tilde{\chi}_{29}\,(\nabla T)^{abcd}\,\varepsilon_{cdef}\,\tilde{R}^{ef}{}_{ab} & \tilde{\chi}_{30}\,(\nabla T)^{abcd}\,\varepsilon_{bdef}\,\tilde{R}^{ef}{}_{ac} \\
\tilde{\chi}_{31}\,(\nabla T)^{abcd}\,\varepsilon_{bcdf}\,\tilde{R}^{ef}{}_{ae} & \tilde{\chi}_{32}\,(\nabla T)^{abcd}\,\varepsilon_{adef}\,\tilde{R}^{ef}{}_{bc} \\
\tilde{\chi}_{33}\,(\nabla T)^{abcd}\,\varepsilon_{acdf}\,\tilde{R}^{ef}{}_{be} & \tilde{\chi}_{34}\,(\nabla T)^{abcd}\,\varepsilon_{abef}\,\tilde{R}^{ef}{}_{cd} \\
\tilde{\chi}_{35}\,(\nabla T)^{abcd}\,\varepsilon_{abdf}\,\tilde{R}^{ef}{}_{ce} & \tilde{\chi}_{36}\,(\nabla T)^{abcd}\,\varepsilon_{abcd}\,\tilde{R}^{ef}{}_{ef}
\end{array} \qquad (31)$$

B.6 $\varepsilon \cdot (\nabla T)^2$ (38 terms)

$$\begin{array}{ll}
\tilde{\xi}_1\,(\nabla T)^{abcd}\,(\nabla T)_{cd}{}^{ef}\,\varepsilon_{abef} & \tilde{\xi}_2\,(\nabla T)^{abcd}\,(\nabla T)_{c}{}^{e}{}_{d}{}^{f}\,\varepsilon_{abef} \\
\tilde{\xi}_3\,(\nabla T)^{abcd}\,(\nabla T)^{e}{}_{cd}{}^{f}\,\varepsilon_{abef} & \tilde{\xi}_4\,(\nabla T)^{abcd}\,(\nabla T)^{ef}{}_{cd}\,\varepsilon_{abef} \\
\tilde{\xi}_5\,(\nabla T)^{ab}{}_{b}{}^{c}\,(\nabla T)^{de}{}_{e}{}^{f}\,\varepsilon_{acdf} & \tilde{\xi}_6\,(\nabla T)^{abcd}\,(\nabla T)_{bc}{}^{ef}\,\varepsilon_{adef} \\
\tilde{\xi}_7\,(\nabla T)^{abcd}\,(\nabla T)_{b}{}^{e}{}_{c}{}^{f}\,\varepsilon_{adef} & \tilde{\xi}_8\,(\nabla T)^{abcd}\,(\nabla T)_{cb}{}^{ef}\,\varepsilon_{adef} \\
\tilde{\xi}_9\,(\nabla T)^{ab}{}_{b}{}^{c}\,(\nabla T)_{c}{}^{def}\,\varepsilon_{adef} & \tilde{\xi}_{10}\,(\nabla T)^{abcd}\,(\nabla T)_{c}{}^{e}{}_{b}{}^{f}\,\varepsilon_{adef} \\
\tilde{\xi}_{11}\,(\nabla T)^{ab}{}_{b}{}^{c}\,(\nabla T)^{d}{}_{c}{}^{ef}\,\varepsilon_{adef} & \tilde{\xi}_{12}\,(\nabla T)^{ab}{}_{b}{}^{c}\,(\nabla T)^{de}{}_{c}{}^{f}\,\varepsilon_{adef} \\
\tilde{\xi}_{13}\,(\nabla T)^{abcd}\,(\nabla T)^{e}{}_{bc}{}^{f}\,\varepsilon_{adef} & \tilde{\xi}_{14}\,(\nabla T)^{abcd}\,(\nabla T)^{e}{}_{cb}{}^{f}\,\varepsilon_{adef} \\
\tilde{\xi}_{15}\,(\nabla T)^{a}{}_{a}{}^{bc}\,(\nabla T)^{de}{}_{e}{}^{f}\,\varepsilon_{bcdf} & \tilde{\xi}_{16}\,(\nabla T)^{ab}{}_{a}{}^{c}\,(\nabla T)^{de}{}_{e}{}^{f}\,\varepsilon_{bcdf} \\
\tilde{\xi}_{17}\,(\nabla T)^{a}{}_{a}{}^{bc}\,(\nabla T)^{d}{}_{d}{}^{ef}\,\varepsilon_{bcef} & \tilde{\xi}_{18}\,(\nabla T)^{a}{}_{a}{}^{bc}\,(\nabla T)^{de}{}_{d}{}^{f}\,\varepsilon_{bcef} \\
\tilde{\xi}_{19}\,(\nabla T)^{ab}{}_{a}{}^{c}\,(\nabla T)^{de}{}_{d}{}^{f}\,\varepsilon_{bcef} & \tilde{\xi}_{20}\,(\nabla T)_{ac}{}^{ef}\,(\nabla T)^{abcd}\,\varepsilon_{bdef} \\
\tilde{\xi}_{21}\,(\nabla T)_{a}{}^{e}{}_{c}{}^{f}\,(\nabla T)^{abcd}\,\varepsilon_{bdef} & \tilde{\xi}_{22}\,(\nabla T)^{ab}{}_{a}{}^{c}\,(\nabla T)_{c}{}^{def}\,\varepsilon_{bdef} \\
\tilde{\xi}_{23}\,(\nabla T)^{abcd}\,(\nabla T)_{c}{}^{e}{}_{a}{}^{f}\,\varepsilon_{bdef} & \tilde{\xi}_{24}\,(\nabla T)^{ab}{}_{a}{}^{c}\,(\nabla T)^{d}{}_{c}{}^{ef}\,\varepsilon_{bdef} \\
\tilde{\xi}_{25}\,(\nabla T)^{ab}{}_{a}{}^{c}\,(\nabla T)^{de}{}_{c}{}^{f}\,\varepsilon_{bdef} & \tilde{\xi}_{26}\,(\nabla T)_{a}{}^{def}\,(\nabla T)^{ab}{}_{b}{}^{c}\,\varepsilon_{cdef} \\
\tilde{\xi}_{27}\,(\nabla T)_{ab}{}^{ef}\,(\nabla T)^{abcd}\,\varepsilon_{cdef} & \tilde{\xi}_{28}\,(\nabla T)^{abcd}\,(\nabla T)_{ba}{}^{ef}\,\varepsilon_{cdef} \\
\tilde{\xi}_{29}\,(\nabla T)^{a}{}_{a}{}^{bc}\,(\nabla T)_{b}{}^{def}\,\varepsilon_{cdef} & \tilde{\xi}_{30}\,(\nabla T)^{ab}{}_{a}{}^{c}\,(\nabla T)_{b}{}^{def}\,\varepsilon_{cdef} \\
\tilde{\xi}_{31}\,(\nabla T)^{abcd}\,(\nabla T)_{b}{}^{e}{}_{a}{}^{f}\,\varepsilon_{cdef} & \tilde{\xi}_{32}\,(\nabla T)^{ab}{}_{ab}\,(\nabla T)^{cdef}\,\varepsilon_{cdef} \\
\tilde{\xi}_{33}\,(\nabla T)^{ab}{}_{b}{}^{c}\,(\nabla T)^{d}{}_{a}{}^{ef}\,\varepsilon_{cdef} & \tilde{\xi}_{34}\,(\nabla T)^{a}{}_{a}{}^{bc}\,(\nabla T)^{d}{}_{b}{}^{ef}\,\varepsilon_{cdef} \\
\tilde{\xi}_{35}\,(\nabla T)^{ab}{}_{a}{}^{c}\,(\nabla T)^{d}{}_{b}{}^{ef}\,\varepsilon_{cdef} & \tilde{\xi}_{36}\,(\nabla T)^{ab}{}_{b}{}^{c}\,(\nabla T)^{de}{}_{a}{}^{f}\,\varepsilon_{cdef} \\
\tilde{\xi}_{37}\,(\nabla T)^{a}{}_{a}{}^{bc}\,(\nabla T)^{de}{}_{b}{}^{f}\,\varepsilon_{cdef} & \tilde{\xi}_{38}\,(\nabla T)^{ab}{}_{a}{}^{c}\,(\nabla T)^{de}{}_{b}{}^{f}\,\varepsilon_{cdef}
\end{array} \tag{32}$$

B.7 $\varepsilon \cdot \tilde{R} \times F^2$ (12 terms)

$$\begin{array}{ll}
\tilde{\lambda}_1\,\varepsilon_{cdef}\,F^{ab}\,F^{cd}\,\tilde{R}_{ab}{}^{ef} & \tilde{\lambda}_2\,\varepsilon_{bdef}\,F^{ab}\,F^{cd}\,\tilde{R}_{ac}{}^{ef} \\
\tilde{\lambda}_3\,\varepsilon_{cdef}\,F^{ab}\,F^{cd}\,\tilde{R}_{a}{}^{e}{}_{b}{}^{f} & \tilde{\lambda}_4\,\varepsilon_{bdef}\,F^{ab}\,F^{cd}\,\tilde{R}_{a}{}^{e}{}_{c}{}^{f} \\
\tilde{\lambda}_5\,\varepsilon_{bcdf}\,F^{ab}\,F^{cd}\,\tilde{R}_{a}{}^{e}{}_{e}{}^{f} & \tilde{\lambda}_6\,\varepsilon_{cdef}\,F_{a}{}^{c}\,F^{ab}\,\tilde{R}_{b}{}^{def} \\
\tilde{\lambda}_7\,\varepsilon_{cdef}\,F_{ab}\,F^{ab}\,\tilde{R}^{cdef} & \tilde{\lambda}_8\,\varepsilon_{cdef}\,F_{a}{}^{c}\,F^{ab}\,\tilde{R}^{de}{}_{b}{}^{f} \\
\tilde{\lambda}_9\,\varepsilon_{cdef}\,F^{ab}\,F^{cd}\,\tilde{R}^{ef}{}_{ab} & \tilde{\lambda}_{10}\,\varepsilon_{bdef}\,F^{ab}\,F^{cd}\,\tilde{R}^{ef}{}_{ac} \\
\tilde{\lambda}_{11}\,\varepsilon_{bcdf}\,F^{ab}\,F^{cd}\,\tilde{R}^{ef}{}_{ae} & \tilde{\lambda}_{12}\,\varepsilon_{abcd}\,F^{ab}\,F^{cd}\,\tilde{R}^{ef}{}_{ef}
\end{array} \tag{33}$$

B.8 $\varepsilon \cdot T^2 \times F$ (15 terms)

$$
\begin{array}{ll}
\tilde{\kappa}_1\, \varepsilon_{bdef}\, F^{ab}\, T_a{}^{cd}\, T_c{}^{ef} & \tilde{\kappa}_2\, \varepsilon_{bdef}\, F^{ab}\, T_c{}^{ef}\, T^c{}_a{}^d \\
\tilde{\kappa}_3\, \varepsilon_{abcf}\, F^{ab}\, T^{cde}\, T_{de}{}^f & \tilde{\kappa}_4\, \varepsilon_{bcef}\, F^{ab}\, T^c{}_a{}^d\, T_d{}^{ef} \\
\tilde{\kappa}_5\, \varepsilon_{abef}\, F^{ab}\, T^c{}_c{}^d\, T_d{}^{ef} & \tilde{\kappa}_6\, \varepsilon_{cdef}\, F^{ab}\, T_{ab}{}^c\, T^{def} \\
\tilde{\kappa}_7\, \varepsilon_{cdef}\, F^{ab}\, T^c{}_{ab}\, T^{def} & \tilde{\kappa}_8\, \varepsilon_{bdef}\, F^{ab}\, T^c{}_{ac}\, T^{def} \\
\tilde{\kappa}_9\, \varepsilon_{cdef}\, F^{ab}\, T_a{}^{cd}\, T^e{}_b{}^f & \tilde{\kappa}_{10}\, \varepsilon_{bdef}\, F^{ab}\, T_a{}^{cd}\, T^e{}_c{}^f \\
\tilde{\kappa}_{11}\, \varepsilon_{bdef}\, F^{ab}\, T^c{}_a{}^d\, T^e{}_c{}^f & \tilde{\kappa}_{12}\, \varepsilon_{bcef}\, F^{ab}\, T^c{}_a{}^d\, T^e{}_d{}^f \\
\tilde{\kappa}_{13}\, \varepsilon_{abef}\, F^{ab}\, T^c{}_c{}^d\, T^e{}_d{}^f & \tilde{\kappa}_{14}\, \varepsilon_{bcdf}\, F^{ab}\, T_a{}^{cd}\, T^e{}_e{}^f \\
\tilde{\kappa}_{15}\, \varepsilon_{bcdf}\, F^{ab}\, T^c{}_a{}^d\, T^e{}_e{}^f &
\end{array}
\tag{34}
$$

B.9 $\varepsilon \cdot F^2 \times \nabla T$ (18 terms)

$$
\begin{array}{ll}
\tilde{\mu}_1\, (\nabla T)^{abcd}\, \varepsilon_{cdef}\, F_a{}^e\, F_b{}^f & \tilde{\mu}_2\, (\nabla T)^{abcd}\, \varepsilon_{bdef}\, F_a{}^e\, F_c{}^f \\
\tilde{\mu}_3\, (\nabla T)^{abcd}\, \varepsilon_{adef}\, F_b{}^e\, F_c{}^f & \tilde{\mu}_4\, (\nabla T)^{abcd}\, \varepsilon_{abef}\, F_c{}^e\, F_d{}^f \\
\tilde{\mu}_5\, (\nabla T)^{abcd}\, \varepsilon_{bcdf}\, F_a{}^e\, F_e{}^f & \tilde{\mu}_6\, (\nabla T)^{abcd}\, \varepsilon_{acdf}\, F_b{}^e\, F_e{}^f \\
\tilde{\mu}_7\, (\nabla T)^{abcd}\, \varepsilon_{abdf}\, F_c{}^e\, F_e{}^f & \tilde{\mu}_8\, (\nabla T)^{abcd}\, \varepsilon_{cdef}\, F_{ab}\, F^{ef} \\
\tilde{\mu}_9\, (\nabla T)^{abcd}\, \varepsilon_{bdef}\, F_{ac}\, F^{ef} & \tilde{\mu}_{10}\, (\nabla T)^{ab}{}_b{}^c\, \varepsilon_{cdef}\, F_a{}^d\, F^{ef} \\
\tilde{\mu}_{11}\, (\nabla T)^{abcd}\, \varepsilon_{adef}\, F_{bc}\, F^{ef} & \tilde{\mu}_{12}\, (\nabla T)^a{}_a{}^{bc}\, \varepsilon_{cdef}\, F_b{}^d\, F^{ef} \\
\tilde{\mu}_{13}\, (\nabla T)^{ab}{}_a{}^c\, \varepsilon_{cdef}\, F_b{}^d\, F^{ef} & \tilde{\mu}_{14}\, (\nabla T)^{abcd}\, \varepsilon_{abef}\, F_{cd}\, F^{ef} \\
\tilde{\mu}_{15}\, (\nabla T)^{ab}{}_b{}^c\, \varepsilon_{adef}\, F_c{}^d\, F^{ef} & \tilde{\mu}_{16}\, (\nabla T)^{ab}{}_a{}^c\, \varepsilon_{bdef}\, F_c{}^d\, F^{ef} \\
\tilde{\mu}_{17}\, (\nabla T)^{ab}{}_{ab}\, \varepsilon_{cdef}\, F^{cd}\, F^{ef} & \tilde{\mu}_{18}\, (\nabla T)^{abcd}\, \varepsilon_{abcd}\, F_{ef}\, F^{ef}
\end{array}
\tag{35}
$$

References

[1] W. Barker, E. Sherrill, A. Sherrill, and M. P. Sherrill, "Conformal Poincaré gauge theory: a new spin on vector torsion," arXiv:2406.12826.

[2] I. L. Shapiro, "Physical aspects of the space-time torsion," Phys. Rept. **357**, 113 (2002), arXiv:hep-th/0103093.

[3] Yu. N. Obukhov, D. Puetzfeld, and F. W. Hehl, "New aspects of the gravitational spin–torsion coupling," arXiv:2410.01355.

[4] G. K. Karananas, "The particle spectrum of parity-violating Poincaré gravitational theory," Class. Quant. Grav. **32**, 055012 (2015).

[5] M. Blagojević and B. Cvetković, "General Poincaré gauge theory: Hamiltonian structure and particle spectrum," Phys. Rev. D **98**, 104018 (2018).

[6] K. Hayashi and T. Shirafuji, "New general relativity," Phys. Rev. D **19**, 3524 (1979).

[7] S. Holst, "Barbero's Hamiltonian derived from a generalized Hilbert–Palatini action," Phys. Rev. D **53**, 5966 (1996), arXiv:gr-qc/9511026.

[8] H. T. Nieh and M. L. Yan, "An identity in Riemann–Cartan geometry," J. Math. Phys. **23**, 373 (1982).

[9] T. Nutma, "`xTras`: A field-theory inspired `xAct` package for Mathematica," Comput. Phys. Commun. **185**, 1719 (2014), arXiv:1308.3493.

[10] V. Nikiforova, S. Randjbar-Daemi and V. Rubakov, "Infrared Modified Gravity with Dynamical Torsion," Phys. Rev. D **80**, 124050 (2009), arXiv:0905.3732.

[11] K. Aoki and K. Shimada, "Galileon and generalized Galileon with projective invariance in a metric-affine formalism," Phys. Rev. D **101**, 124023 (2020), arXiv:2009.11739.

[12] Y. Itin, "Maxwell's field coupled nonminimally to quadratic torsion: induced axion field and birefringence of the vacuum," Phys. Rev. D **68**, 127701 (2003), arXiv:gr-qc/0307063.

[13] S. Bahamonde et al., "Cosmology of Cubic Poincaré Gauge gravity," arXiv:2506.17017 (2025).